

(Webscrapping & Sentiment Analysis) Movie Overview from Themoviedb.org

Themoviedb.org is a popular website that aggregates movie overviews, reviews and ratings for movies and TV shows. It provides a comprehensive overview of the critical and audience reception of various entertainment content

- ▼ a. Key features: movie's basic information (movie id, title, release date, vote average, overview, runtime, box office earning)
- b. The collected dataset spans a period of 5 months from 1st January 2025 to 1st June 2025

GOAL: Explore Themoviedb.org to discover the trending movie and the sentiment its movie overviews

1. Crawl data using API of TMDB (Themoviedb.org)

```
[92... import requests
import pandas as pd

API_KEY = "8ebc019c041f4ba9fdc43675eb228764"
BASE_URL = "https://api.themoviedb.org/3/discover/movie"
MOVIE_URL = "https://api.themoviedb.org/3/movie/"

movies = []
page = 1
total_pages = 30

while page <= total_pages:
    params = {
        "api_key": API_KEY,
        "language": "en-US",
        "sort_by": "popularity.desc",
        "primary_release_date.gte": "2025-01-01",
        "primary_release_date.lte": "2025-06-01",
        "page": page,
    }
    #Send request to API
    response = requests.get(BASE_URL, params=params)
    data = response.json()

    print(f"Page {page} - Movies found: {len(data.get('results', []))}")

    for movie in data.get("results", []):
        # Get information about each movie
        movie_id = movie["id"]
        movie_details = requests.get(f"{MOVIE_URL}{movie_id}", params={"api_key": API_KEY, "language":

        box_office_earnings = movie_details.get("revenue", 0) # Box office earnings
        runtime = movie_details.get("runtime", 0) # Runtime
```

```

movies.append({
    "id": movie["id"],
    "title": movie["title"],
    "original_title": movie.get("original_title", ""),
    "release_date": movie.get("release_date", ""),
    "popularity": movie.get("popularity", 0),
    "vote_average": movie.get("vote_average", 0),
    "vote_count": movie.get("vote_count", 0),
    "overview": movie.get("overview", ""),
    "runtime": runtime,
    "box_office_earnings": box_office_earnings
})

```

page += 1

Page 1 – Movies found: 20
 Page 2 – Movies found: 20
 Page 3 – Movies found: 20
 Page 4 – Movies found: 20
 Page 5 – Movies found: 20
 Page 6 – Movies found: 20
 Page 7 – Movies found: 20
 Page 8 – Movies found: 20
 Page 9 – Movies found: 20
 Page 10 – Movies found: 20
 Page 11 – Movies found: 20
 Page 12 – Movies found: 20
 Page 13 – Movies found: 20
 Page 14 – Movies found: 20
 Page 15 – Movies found: 20
 Page 16 – Movies found: 20
 Page 17 – Movies found: 20
 Page 18 – Movies found: 20
 Page 19 – Movies found: 20
 Page 20 – Movies found: 20
 Page 21 – Movies found: 20
 Page 22 – Movies found: 20
 Page 23 – Movies found: 20
 Page 24 – Movies found: 20
 Page 25 – Movies found: 20
 Page 26 – Movies found: 20
 Page 27 – Movies found: 20
 Page 28 – Movies found: 20
 Page 29 – Movies found: 20
 Page 30 – Movies found: 20

	id	title	original_title	release_date	popularity \
0	870028	The Accountant ²	The Accountant ²	2025-04-23	585.1611
1	552524	Lilo & Stitch	Lilo & Stitch	2025-05-17	457.1902
2	1315988	Mikaela	Mikaela	2025-01-31	434.0803
3	1233413	Sinners	Sinners	2025-04-16	333.1718
4	1087891	The Amateur	The Amateur	2025-04-09	454.4510

	vote_average	vote_count \
0	7.200	670
1	7.100	605
2	6.186	43
3	7.554	1386
4	6.804	369

	overview	runtime \
0	When an old acquaintance is murdered, Wolff is...	133
1	The wildly funny and touching story of a lonel...	108
2	During the eve of the 6th of January, a record...	90
3	Trying to leave their troubled lives behind, t...	138
4	After his life is turned upside down when his ...	123

```
box_office_earnings
0      102116542
1      807600000
2           0
3      357628506
4      95808586
```

```
[94... df_movies = pd.DataFrame(movies)
df_movies
```

[94...		id	title	original_title	release_date	popularity	vote_average	vote_count	overview	runt
	0	870028	The Accountant ²	The Accountant ²	2025-04-23	585.1611	7.200	670	When an old acquaintance is murdered, Wolff is...	
	1	552524	Lilo & Stitch	Lilo & Stitch	2025-05-17	457.1902	7.100	605	The wildly funny and touching story of a lonel...	
	2	1315988	Mikaela	Mikaela	2025-01-31	434.0803	6.186	43	During the eve of the 6th of January, a record...	
	3	1233413	Sinners	Sinners	2025-04-16	333.1718	7.554	1386	Trying to leave their troubled lives behind, t...	
	4	1087891	The Amateur	The Amateur	2025-04-09	454.4510	6.804	369	After his life is turned upside down when his ...	
	...	...	...	...	...	...	...	...	...	
	595	1414052	Mr.Robot	미스터 로봇	2025-04-04	1.0231	5.000	2	A futuristic world where robots have become po...	
	596	1465026	Untold: The Fall of Favre	Untold: The Fall of Favre	2025-05-19	1.2069	5.583	12	This eye-opening documentary delves into Brett...	
	597	1176529	Everything About My Wife	Everything About My Wife	2025-02-26	1.0088	1.000	1	The film revolves around an unhappy married ma...	
	598	1414275	Diddy: Monster's Fall	Diddy: Monster's Fall	2025-01-10	1.0671	6.500	2	Disgraced hip-hop mogul, Sean 'Diddy' Combs co...	
	599	1441461	Kolektor	Kolektor	2025-03-07	1.2967	0.000	0	Jessie, a single man trying to look for love. ...	

Convert the variable "release_date" which is under the form of date into the numeric variables named "Days_since_release"

[97...	df_movies['release_date'] = pd.to_datetime(df_movies['release_date'])									
[99...	reference_date = pd.to_datetime('2025-06-13')									
[10...	df_movies['days_since_release'] = (reference_date-df_movies['release_date']).dt.days									
[10...										
		id	title	original_title	release_date	popularity	vote_average	vote_count	overview	runtime
	0	870028	The Accountant ²	The Accountant ²	2025-04-23	585.1611	7.200	670	When an old acquaintance is murdered, Wolff is...	
	1	552524	Lilo & Stitch	Lilo & Stitch	2025-05-17	457.1902	7.100	605	The wildly funny and touching story of a lonel...	
	2	1315988	Mikaela	Mikaela	2025-01-31	434.0803	6.186	43	During the eve of the 6th of January, a record...	
	3	1233413	Sinners	Sinners	2025-04-16	333.1718	7.554	1386	Trying to leave their troubled lives behind, t...	
	4	1087891	The Amateur	The Amateur	2025-04-09	454.4510	6.804	369	After his life is turned upside down when his ...	
	...	...	...	...	...	...	...	...	...	
	595	1414052	Mr.Robot	미스터 로봇	2025-04-04	1.0231	5.000	2	A futuristic world where robots have become po...	
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	599	1441461	Kolektor	Kolektor	2025-03-07	1.2967	0.000	0	Jessie, a single man trying to look for love. ...	

2. Sentiment Analysis of Film Overview

```
[10...] from nltk.sentiment.vader import SentimentIntensityAnalyzer

[10...] import nltk
nltk.download('vader_lexicon')

[nltk_data] Downloading package vader_lexicon to
[nltk_data] /Users/harley.le/nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!

[10...] True

[10...] vader = SentimentIntensityAnalyzer()

[10...] df_movies["overview"]

[10...] 0      When an old acquaintance is murdered, Wolff is...
      1      The wildly funny and touching story of a lonel...
      2      During the eve of the 6th of January, a record...
      3      Trying to leave their troubled lives behind, t...
      4      After his life is turned upside down when his ...
      ...
      595     A futuristic world where robots have become po...
      596     This eye-opening documentary delves into Brett...
      597     The film revolves around an unhappy married ma...
      598     Disgraced hip-hop mogul, Sean 'Diddy' Combs co...
      599     Jessie, a single man trying to look for love. ...
      Name: overview, Length: 600, dtype: object

[11...] overview_list = df_movies["overview"].tolist()

[11...] overview_list

[11...] ['When an old acquaintance is murdered, Wolff is compelled to solve the case. Realizing more extreme measures are necessary, Wolff recruits his estranged and highly lethal brother, Brax, to help. In partnership with Marybeth Medina, they uncover a deadly conspiracy, becoming targets of a ruthless network of killers who will stop at nothing to keep their secrets buried.',
      'The wildly funny and touching story of a lonely Hawaiian girl and the fugitive alien who helps to mend her broken family.',
      'During the eve of the 6th of January, a record-breaking snowstorm sweeps across Spain. In the midst of its chaos, a group of robbers seizes the opportunity to hijack an armoured van. A few meters away is Leo, a finished policeman who has nothing to lose. With the unexpected aid of a young woman, he will try to stop the band from running away with their loot.',
      'Trying to leave their troubled lives behind, twin brothers return to their hometown to start again, only to discover that an even greater evil is waiting to welcome them back.',
      'After his life is turned upside down when his wife is killed in a London terrorist attack, a brilliant but introverted CIA decoder takes matters into his own hands when his supervisors refuse to take action.',
      'Four misfits find themselves struggling with ordinary problems when they are suddenly pulled through a mysterious portal into the Overworld: a bizarre, cubic wonderland that thrives on imagination. To get back home, they'll have to master this world while embarking on a magical quest with an unexpected, expert crafter, Steve.',

[11...] overview_list[0]

[11...] 'When an old acquaintance is murdered, Wolff is compelled to solve the case. Realizing more extreme measures are necessary, Wolff recruits his estranged and highly lethal brother, Brax, to help. In partnership with Marybeth Medina, they uncover a deadly conspiracy, becoming targets of a ruthless network of killers who will stop at nothing to keep their secrets buried.'

[11...] vader.polarity_scores(overview_list[0])

[11...] {'neg': 0.207, 'neu': 0.71, 'pos': 0.083, 'compound': -0.891}

[11...] vader.polarity_scores(overview_list[0])["compound"]

[11...] -0.891
```

```
[12...] sentiments = [vader.polarity_scores(o) ["compound"] for o in overview_list]
```

```
[12...] sentiments
```

```
-0.2144,  
0.7506,  
0.6597,  
-0.1531,  
-0.4767,  
0.885,  
-0.6597,  
-0.5106,  
-0.6705,  
-0.802,  
0.2382,  
-0.8519,  
-0.8442,  
-0.7717,  
-0.4019,  
-0.5719,  
-0.6369,  
0.0,  
0.0,  
-0.8957,
```

```
[12...] df_movies["Overview_sentiment"] = sentiments
```

```
[27...] df_movies
```

```
[27...]      id      title  original_title  release_date  popularity  vote_average  vote_count  overview  runt
```

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...	...	...	...	...	...	...	...	...
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